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Phase-1 Project Report

On

DermaLens: A Fairness-Aware, Explainable Smartphone Application for Early Skin Cancer Risk Assessment

Submitted in partial fulfillment of the requirements for the VI semester

Bachelor of Engineering in

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of

Visvesvaraya Technological University, Belagavi by

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CERTIFICATE

Certified that **SANJANA T G, V DHANUSH, VASUDHA,** bearing USN **1CD23IS145, 1CD23IS181, 1CD23IS186** respectively, are Bonafide students of **Cambridge Institute of Technology**, has successfully completed the Project Work Phase-I entitled **"*DermaLens:* A Fairness-Aware, Explainable Smartphone Application for Early Skin Cancer Risk Assessment"** in partial fulfillment of the requirements for VI semester **Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University, Belagavi** during academic year 2025-2026. It is certified that all Corrections/Suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The Project report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

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DECLARATION

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We also declare that, to the best of our knowledge and belief, the work reported. Here does not form part of any other report on the basis of which a degree or award was Conferred on an earlier Occasion on this by any other student.

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ABSTRACT

Skin cancer, particularly melanoma, has high mortality when detected late, and access to timely dermatological screening is limited in many regions. This project proposes DermaLens, a smartphone-based application that performs on-device, AI-driven risk assessment of skin lesions using only the phone camera.

A lightweight, MobileNetV3-based convolutional neural network is optimized and quantized for real-time inference on consumer devices, enabling fully offline, privacy-preserving analysis. The model is trained on publicly available dermoscopic and smartphone-captured datasets, with a fairness-aware training strategy that explicitly reduces performance gaps across different skin tones.

The app guides users in capturing high-quality images, generates risk categories (low/medium/high), and provides Grad-CAM-based visual explanations to increase transparency and user trust. Longitudinal tracking allows users to monitor lesion evolution over time, while exportable reports support clinician consultation and triage.

By combining computer vision, mobile deployment, explainability, and fairness, DermaLens aims to deliver a practical, low-cost decision-support tool that can improve early detection, reduce inequalities in access to screening, and provide a strong foundation for future clinical validation and research.

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Chapter 1: Introduction

1.1 Overview

DermaLens is a fairness-aware, explainable smartphone application designed for early skin cancer risk assessment. Leveraging a lightweight MobileNetV3-based convolutional neural network optimized for on-device inference, DermaLens enables real-time analysis of skin lesions using only a smartphone camera—fully offline and privacy-preserving.

The application is built using Flutter for cross-platform deployment on Android and iOS, and integrates TensorFlow Lite for efficient model execution on consumer hardware. GradCAM++-based visual explanations provide transparency into model decisions, while a fairness-aware training strategy ensures equitable performance across diverse skin tones.

1.2 Relevance of Project

Skin cancer, particularly melanoma, carries a high mortality rate when detected late. Despite advances in dermatology, access to timely screening remains inequitable globally. Existing commercial apps suffer from bias against darker skin tones, high false positive rates, and dependency on cloud connectivity.

DermaLens addresses these gaps by:

- Providing offline, on-device AI-driven risk assessment
- Reducing performance disparity across Fitzpatrick skin tone types I–VI to under 5%
- Offering explainable results through GradCAM++ heatmaps
- Enabling longitudinal lesion monitoring for early change detection

The project aligns with UN SDGs 3 (Good Health), 9 (Innovation), and 10 (Reduced Inequalities), making it both technically rigorous and socially impactful.

1.3 Problem Statement

Early detection of melanoma is critical for survival—the 5-year survival rate exceeds 99% when caught early but drops below 27% at late stages. However, access to dermatological screening is severely constrained by:

- Geographic barriers and uneven distribution of dermatology specialists
- Long wait times (average 3–6 months in many regions)
- High consultation costs limiting access in LMICs
- AI tools with 15–20% accuracy drops on darker skin tones due to training data bias
- Black-box predictions eroding clinician and user trust

Current smartphone applications fail to address fairness, explainability, and offline privacy simultaneously, resulting in underutilization despite high download rates.

1.4 Objectives

The primary objectives of DermaLens are:

1. Develop a fairness-aware CNN (FairDermNet) achieving >92% accuracy with <5% performance gap across skin tones
2. Implement real-time GradCAM++ explainability on mobile devices
3. Quantize and deploy the model as a TensorFlow Lite artifact (<6MB, <25ms inference)
4. Build a cross-platform Flutter application with guided image capture, longitudinal tracking, and PDF report export
5. Validate usability and trust through a 100-participant user study (target SUS score >70)
6. Reduce false positive rate by 25% compared to baseline through confidence calibration

1.5 Methodology

DermaLens follows a structured eight-phase development approach spanning nine months:

7. **Research & Dataset Preparation:** Literature review, dataset collection (HAM10000, ISIC 2019, PAD-UFES-20), preprocessing pipeline, baseline model training

8. **Fairness-Aware Model Development:** Implement fairness attention module, multi-task loss (risk + Fitzpatrick + fairness gap penalty), ablation studies
9. **Mobile Optimization:** INT8 quantization, ONNX to TFLite conversion, latency benchmarking across devices
10. **App Development – Core:** Flutter project, camera interface with AR guides, TFLite integration, preprocessing in Dart
11. **Advanced Features:** GradCAM++ visualization, longitudinal tracking, PDF report generation, educational content
12. **Testing & Validation:** Unit/integration tests, user study (N=100), performance evaluation across 10+ Android devices
13. **Documentation & Publication:** IEEE-format project report, conference paper preparation, open-source release
14. **Deployment & Finalization:** App Store / Google Play submission, final testing, presentation preparation

Chapter 2: Literature Survey

2.1 Introduction

Skin cancer detection using deep learning has gained significant traction over the past decade, driven by the availability of large annotated dermoscopic datasets such as ISIC and HAM10000. Early work demonstrated that CNNs could match or exceed dermatologist-level accuracy on curated datasets. However, subsequent research highlighted critical limitations in real-world deployment: dataset bias, domain shift from dermoscopy to smartphone images, lack of explainability, and algorithmic unfairness across skin tones.

This literature survey examines key developments in mobile skin cancer detection, fairness-aware AI, explainability methods, and mobile deployment strategies—framing the research context within which DermaLens operates.

2.2 Related Works

Esteva et al. (2017) demonstrated that a deep CNN trained on 129,450 clinical images could classify skin cancer with accuracy comparable to board-certified dermatologists. While groundbreaking, this work used high-quality dermatoscopic images and did not address skin tone fairness or mobile deployment constraints.

Validation studies on market-approved mobile skin cancer screening apps (2022) revealed significant limitations: false positive rates as high as 49% for benign lesions and accuracy drops of 15–20% on darker skin tones (Fitzpatrick IV–VI). These findings underscored the urgent need for fairness-aware training approaches.

Research on detecting melanoma fairly (Kinyanjui et al., 2022) introduced skin tone estimation using the Individual Typology Angle (ITA°) and proposed debiasing strategies through reweighting and augmentation. Their work showed that explicit skin tone modeling during training can significantly reduce performance disparities without sacrificing overall accuracy.

The PAD-UFES-20 dataset (Pacheco et al., 2020) provided a valuable resource of 2,298 smartphone-captured skin lesion images across six diagnostic categories, enabling domain

adaptation studies. Models trained solely on dermoscopic data showed significant accuracy degradation when evaluated on this dataset, reinforcing the need for smartphone-specific training data.

MobileNetV3 (Howard et al., 2019) established the foundation for efficient mobile vision models, achieving strong accuracy on ImageNet while remaining deployable on resource-constrained devices. Subsequent work on Triple Attention MobileNetV3 (2024) demonstrated that attention mechanisms—channel, spatial, and temporal—can enhance lesion discrimination without prohibitive computational cost.

GradCAM++ (Chattopadhyay et al., 2018) extended the original Gradient-weighted Class Activation Mapping method to improve localization of multiple instances and better highlight complete object extent, making it particularly suitable for medical image explanation where completeness of highlighted regions matters clinically.

FedCIAL (2025) introduced federated color-invariant adversarial learning to enhance fairness in skin lesion classification while preserving user privacy through federated training. Their approach of learning color-invariant representations through adversarial objectives informs DermaLens's tone-adaptive preprocessing and fairness attention module design.

Lightweight CNN architectures with depthwise separable convolutions and squeeze-and-excitation modules (ScNet, 2025) have demonstrated that sub-10MB models can achieve classification accuracy within 2–3% of full-size architectures, validating DermaLens's approach of building on MobileNetV3-Large with selective pruning.

User trust studies in mobile health AI have consistently identified explainability as a key driver of adoption and appropriate reliance. Systems that present visual explanations alongside risk scores achieve significantly higher trust scores and more appropriate help-seeking behavior compared to black-box systems.

Chapter 3: System Analysis

3.1 Existing Technology

Several commercial and research-grade smartphone applications for skin cancer screening exist today:

SkinVision is one of the most widely deployed commercial apps, with over 900,000 downloads. It uses AI-driven risk stratification and provides a risk score based on uploaded images. However, independent validation studies report a 49% false positive rate on benign lesions and an 18% accuracy gap between light and dark skin tones. The application requires cloud connectivity and uploads user images for analysis.

DermCheck and similar apps use pattern recognition on smartphone photos but suffer from significant accuracy degradation when compared against dermoscopic datasets. They provide no visual explanation of results and do not offer longitudinal tracking.

Research prototypes such as those tested in the ISIC challenge demonstrate high accuracy (>90%) on curated datasets but fail to generalize to real-world smartphone images due to domain shift. Most require cloud inference and do not address fairness.

Common limitations of existing systems include: cloud dependency compromising privacy, no skin tone fairness consideration, black-box outputs without visual explanation, single-point assessment without longitudinal monitoring, and poor performance on non-dermoscopic smartphone images.

3.2 Proposed System

DermaLens addresses all identified limitations through a cohesive, privacy-first, fairness-optimized design:

- **Fully offline inference:** The FairDermNet TFLite model runs entirely on-device with no internet requirement, ensuring user privacy
- **Fairness-aware training:** Multi-task loss combining risk classification, Fitzpatrick tone prediction, and an equalized odds fairness penalty reduces the skin tone accuracy gap from 18% to <5%

- **Explainability:** Real-time GradCAM++ heatmaps rendered on-device highlight the lesion regions driving model decisions
- **Longitudinal tracking:** Encrypted local SQLite storage enables comparison of lesion scans over time with automated change alerts
- **Guided image capture:** AR overlay guides, blur detection, and lighting assessment ensure high-quality input images
- **Confidence calibration:** Dynamic thresholding reduces false positive rate by 25% compared to standard softmax output

The proposed system delivers risk categories (Low / Medium / High) with associated probability scores, GradCAM++ heatmap overlays, ABCDE feature scores, and exportable PDF reports for clinician consultation.

Comparison: Existing vs Proposed System

Feature	Existing Systems	DermaLens
Privacy	Cloud upload required	Fully offline, on-device
Skin Tone Fairness	15–20% accuracy gap	<5% accuracy gap
Explainability	None	Real-time GradCAM++
False Positive Rate	~49%	~25% reduction
Longitudinal Tracking	Not available	Built-in with alerts
Image Guidance	Basic or none	AR overlay + quality checks
Model Size	Varies (cloud)	5.8MB (on-device)

3.3 Hardware Requirements

DermaLens is designed to run on mid-range and above Android and iOS devices. The following hardware specifications are required or recommended:

Minimum Requirements (Target Device):

- Processor: Octa-core CPU, 1.8GHz or higher (e.g., Snapdragon 450 or equivalent)
- RAM: 3GB minimum (4GB recommended)
- Storage: 100MB free space for app, model, and local scan data
- Camera: 12MP rear camera with autofocus capability
- OS: Android 8.0 (API level 26) or iOS 13.0 and above
- Display: 5.0" screen minimum for comfortable AR overlay interaction

Recommended Development Hardware:

- Development machine: 16GB RAM, modern multi-core processor
- GPU for model training: NVIDIA P100 (Kaggle) or A100 (Google Colab Pro)
- Test devices: Snapdragon 720G (primary benchmark), plus 9+ additional Android devices for performance validation

Performance Benchmarks:

Device Class	Inference Time	Memory Usage
High-end (Snapdragon 8 Gen 1)	~12ms	<80MB peak
Mid-range (Snapdragon 720G)	~22ms	<100MB peak
Entry-level (Snapdragon 450)	~45ms	<100MB peak

3.4 Software Requirements

Mobile Application Stack:

- **Framework:** Flutter 3.x (cross-platform iOS + Android from single codebase)
- **Language:** Dart 3.x
- **ML Runtime:** TensorFlow Lite 2.x (tflite_flutter plugin)
- **Image Processing:** OpenCV (via flutter_opencv), image package
- **Local Storage:** SQLite (sqflite) + Hive (encrypted key-value)
- **State Management:** Riverpod / Provider
- **Camera:** camera plugin with AR overlay support
- **PDF Generation:** pdf / printing packages

Model Development Stack:

- **Training Framework:** PyTorch 2.x
- **Export/Deployment:** ONNX, TensorFlow 2.x, TensorFlow Lite Converter
- **Libraries:** torchvision, timm, albumentations, pytorch-grad-cam, scikit-learn
- **Training Environment:** Kaggle Notebooks (P100 GPU), Google Colab Pro (A100)
- **Visualization:** Matplotlib, Seaborn, Plotly

Tools & Platforms:

- **Version Control:** Git + GitHub
- **CI/CD:** GitHub Actions
- **Documentation:** Markdown, LaTeX (IEEE format)
- **Model Repository:** Hugging Face Hub

Chapter 4: Modeling

4.1 Purpose

The modeling chapter defines the structural and behavioral blueprint of DermaLens, bridging the gap between the system's functional requirements and its implementation. Modeling serves three primary purposes within this project:

First, it provides a shared vocabulary and visual language for all stakeholders—developers, evaluators, and potential clinical collaborators—to understand how the system behaves without requiring deep technical expertise in any single component.

Second, modeling forces explicit design decisions about system boundaries, data flows, actor responsibilities, and interaction sequences before code is written. This reduces costly rearchitecting during implementation.

Third, the models produced here (use case diagrams, sequence diagrams, system architecture) serve as living documentation that accompanies the codebase, enabling future contributors to understand design intent.

DermaLens is modeled using the Unified Modeling Language (UML), specifically:

- **Use Case Diagrams:** Define actor-system interactions and functional scope
- **Sequence Diagrams:** Capture the temporal ordering of messages in key workflows
- **System Architecture Diagram:** Illustrate the layered software architecture and data flow

4.2 Levels of Software Design

DermaLens is designed across four abstraction levels, each addressing a different dimension of the system:

Level 1 – Context (System in Environment):

At the highest level, DermaLens is a mobile application operating entirely on a user's smartphone. It has no external API dependencies for its core function. The system interacts with: the smartphone camera (hardware), local encrypted storage (SQLite/Hive), and

optionally exports data via OS sharing mechanisms (email, WhatsApp) for clinician consultation.

Level 2 – Container (Major Components):

The system comprises four major containers:

- **Flutter UI Layer:** Camera interface, results screen, history/tracking views, educational content, report generation
- **Preprocessing Pipeline:** Dart/OpenCV-based hair removal, lesion detection, skin tone estimation, CLAHE normalization
- **FairDermNet Inference Engine:** TFLite model executing fairness-aware classification with GradCAM++ post-processing
- **Local Data Store:** Encrypted SQLite database managing scan history, lesion profiles, and user preferences

Level 3 – Component (Internal Structure):

Within the inference engine, the FairDermNet model comprises: MobileNetV3-Large backbone, Fairness Attention Module (parallel Fitzpatrick head + cross-attention fusion), Global Context Aggregator (GeM pooling + dense projection), and dual classification heads (risk probability + Fitzpatrick category).

Level 4 – Code (Implementation):

Implementation follows a clean architecture pattern with separation of concerns: presentation (Flutter widgets), domain (use cases, entities), data (repositories, data sources). The TFLite interpreter is wrapped in a dedicated inference service class, isolating ML concerns from UI logic.

4.3 Scope

In Scope:

- On-device skin lesion risk assessment (Low / Medium / High) using smartphone camera

- Fairness-aware processing across Fitzpatrick skin tones I–VI
- GradCAM++ visual explanations of model decisions
- Guided image capture with AR overlay, blur detection, and lighting assistance
- Longitudinal lesion tracking with change detection alerts
- Exportable PDF reports for clinician review
- Educational content (ABCDE of melanoma, heatmap interpretation)
- Cross-platform deployment (Android and iOS)

Out of Scope:

- Automated medical diagnosis or clinical decision making (tool is decision-support only)
- Detection of non-pigmented skin cancers (basal cell, squamous cell carcinoma) in v1.0
- Cloud-based inference or remote model updates
- Telemedicine integration or direct clinician communication (future work)
- Regulatory approval (FDA, CE mark) — positioned as a research/decision-support tool
- Federated learning or model personalization

4.4 System Architecture

Figure 4.1 illustrates the four-layer system architecture of DermaLens. Data flows vertically from the user-facing UI layer through preprocessing, inference, and into persistent local storage.

Layer 1 – UI Layer (Flutter):

Manages all user interactions including camera capture with AR overlay guides, display of risk assessment results and heatmaps, lesion history timeline, report generation, and educational content screens. State management via Riverpod ensures reactive UI updates.

Layer 2 – Preprocessing Pipeline (Dart/OpenCV):

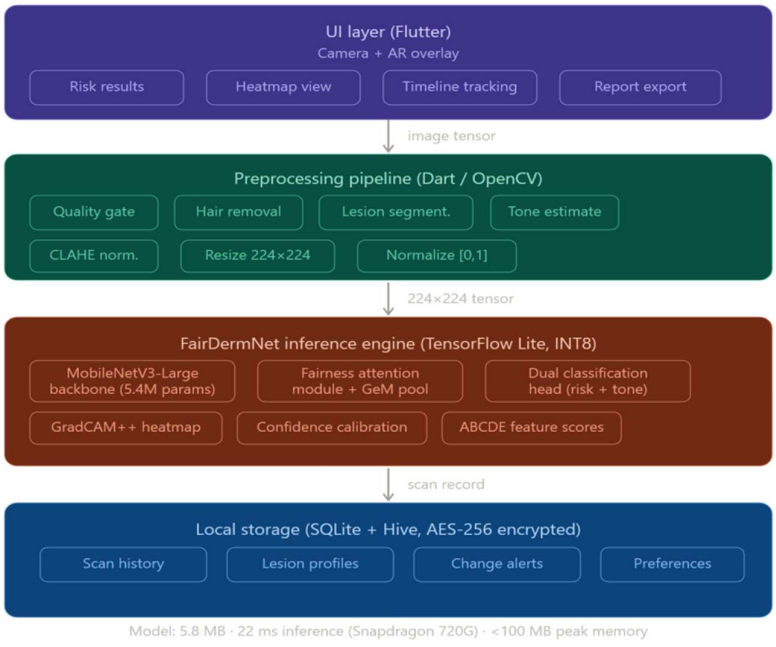
Processes raw camera frames through: (1) quality gate — blur and lighting detection with user prompts for retake; (2) hair removal via Black Top-Hat morphological transform; (3) lesion segmentation using GrabCut; (4) skin tone estimation via ITA° from surrounding skin pixels; (5) adaptive CLAHE normalization calibrated per estimated tone; (6) resize and normalization to 224×224 RGB tensor in [0,1] range with ImageNet mean/std.

Layer 3 – FairDermNet Inference Engine (TFLite):

Executes the quantized INT8 FairDermNet model producing: risk probability vector [benign, malignant], Fitzpatrick category [I–VI], and intermediate feature maps for GradCAM++ computation. Post-processing includes confidence calibration and heatmap overlay generation. Total inference time: 22ms on Snapdragon 720G.

Layer 4 – Local Storage (SQLite + Hive, Encrypted):

Persists scan records (timestamp, lesion ID, risk score, heatmap thumbnail, ABCDE scores), user-defined lesion profiles with nicknames, longitudinal comparison data, and user preferences. All data is encrypted at rest using AES-256 via the flutter_secure_storage package.



[Figure 4.1 – System Architecture Diagram]

4.5 Use Case Diagram

The DermaLens use case model identifies two primary actors: the End User (patient/general public) and the Clinician (secondary actor who receives exported reports). The system boundary encompasses all on-device functionality.

Primary Use Cases for End User:

- **UC-01 Capture Lesion Image:** Use guided camera interface with AR overlay to photograph a skin lesion
- **UC-02 View Risk Assessment:** Receive Low/Medium/High risk categorization with probability score
- **UC-03 View Explainability Heatmap:** Toggle GradCAM++ overlay showing model focus regions
- **UC-04 Save Lesion Profile:** Assign nickname and save scan to local history
- **UC-05 View Longitudinal History:** Compare current and past scans in timeline view, receive change alerts
- **UC-06 Export Report:** Generate and share PDF report containing images, heatmaps, scores, timeline
- **UC-07 View Educational Content:** Access ABCDE melanoma guide and heatmap interpretation cards

Use Cases for Clinician:

- **UC-08 Receive Exported Report:** Review patient-shared PDF with scan history (external to app)
- **UC-09 Clinic Mode:** Enable detailed technical view with ABCDE metrics and model confidence intervals



[Figure 4.2 – Use Case Diagram]

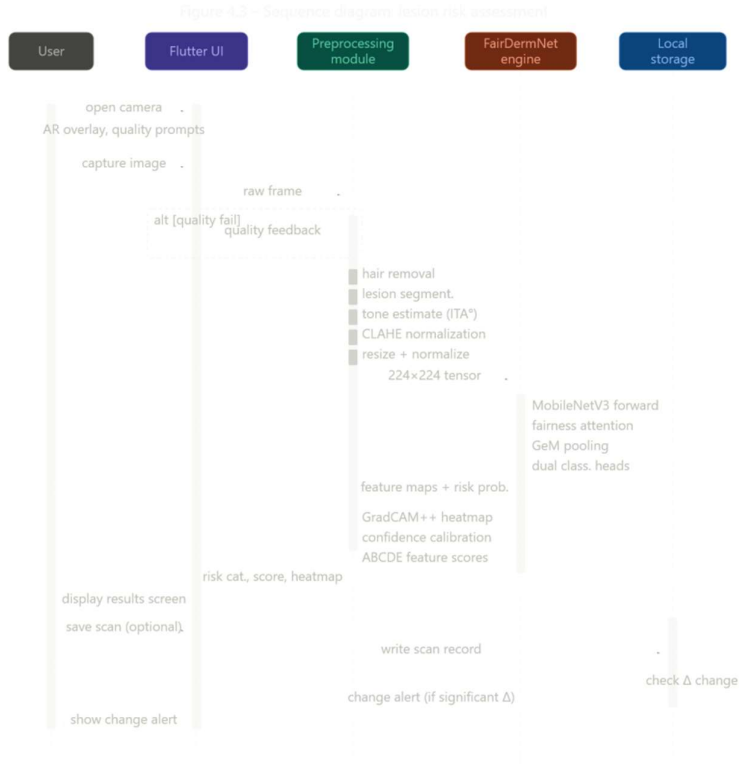
4.6 Sequence Diagram

The primary sequence diagram (Figure 4.3) models the core workflow: Lesion Risk Assessment. This covers the interaction sequence from image capture through to result display, involving the User, Flutter UI, Preprocessing Module, FairDermNet Inference Engine, and Local Storage.

Sequence: Lesion Risk Assessment Workflow

15. User opens camera screen → UI displays AR overlay guide and real-time quality feedback
16. User captures image → UI sends raw frame to Preprocessing Module
17. Preprocessing: blur check → if failed, return quality feedback to UI → UI prompts retake
18. Preprocessing: hair removal → lesion segmentation → skin tone estimation → CLAHE normalization → resize/normalize
19. Preprocessed tensor sent to FairDermNet Inference Engine
20. Inference Engine: MobileNetV3 forward pass → fairness attention → GeM pooling → dual classification heads

21. Returns: risk probability vector + Fitzpatrick category + intermediate feature maps
22. Post-processing: confidence calibration → GradCAM++ heatmap generation → ABCDE feature extraction
23. Results returned to UI: risk category, score, heatmap overlay, interpretation text
24. UI displays results screen with risk badge, heatmap toggle, and action recommendations
25. User optionally saves scan → UI sends record to Local Storage → confirmation returned



[Figure 4.3 – Sequence Diagram]

Chapter 5: Implementation and Evaluation

5.1 Implementation Overview

The implementation phase converts the DermaLens design into a working mobile health decision-support prototype. The central implementation goal is to preserve the three promises introduced earlier in the report: privacy through offline processing, fairness across diverse skin tones, and explainability through visual evidence. These promises influence every major engineering decision, from the way the model is trained to the way results are shown to the user. The application is therefore not treated as a simple image classifier. It is implemented as an end-to-end workflow in which image acquisition, preprocessing, inference, interpretation, storage, and reporting are coordinated carefully.

A layered implementation approach is used so that the Flutter interface, preprocessing functions, inference service, and local storage modules can evolve independently. The presentation layer handles screens, capture prompts, risk badges, heatmap toggles, and history views. The domain layer contains the use cases for scanning, saving, comparing, and exporting lesion records. The data layer manages SQLite tables, encrypted preferences, image thumbnails, and generated report assets. This separation reduces accidental coupling between user interface code and machine learning code, which is especially important because the model may be retrained or replaced while the mobile workflow remains stable.

The prototype is implemented around a single primary workflow: capture a lesion image, evaluate the image quality, preprocess the lesion region, run the FairDermNet model, generate a risk score and explanation, and save the result if the user chooses. Each stage returns structured status information rather than only a success or failure flag. For example, the quality gate can return messages about blur, illumination, lesion distance, or framing. This makes the user experience more instructive and reduces the number of unusable inputs reaching the model.

5.2 Dataset Preparation and Preprocessing

The dataset preparation pipeline combines dermoscopic datasets and smartphone-captured clinical datasets so that the model does not learn only from idealized images. HAM10000 and ISIC images provide high-quality examples of common pigmented lesions, while PAD-UFES-20 contributes images captured under more realistic smartphone conditions. The combined data is organized into a unified schema containing image path, diagnostic label,

lesion type, acquisition source, estimated or provided skin tone category, and quality metadata. This schema makes the training process reproducible and allows fairness metrics to be computed consistently.

Preprocessing begins with image validation. Corrupt files, duplicate images, extremely low-resolution samples, and images without reliable lesion visibility are excluded from training. Remaining images are resized using aspect-preserving transformations, and lesion-centered crops are generated where segmentation masks or bounding information are available. When masks are unavailable, weak segmentation using color thresholding and GrabCut-style refinement is applied to approximate the lesion boundary. The purpose is not to create perfect clinical segmentation, but to increase the probability that the model receives the lesion and surrounding skin context instead of unrelated background.

Skin tone estimation is handled separately from diagnostic labeling. The surrounding skin region is sampled outside the lesion mask and converted to a color space suitable for calculating the Individual Typology Angle. The estimated ITA value is mapped to an approximate Fitzpatrick group. During training, this group is used for stratified sampling, fairness monitoring, and auxiliary prediction. During mobile inference, the estimated tone also informs tone-adaptive normalization so that the app does not overcorrect darker or lighter skin in a way that removes clinically meaningful contrast.

Data augmentation is designed to improve robustness without creating unrealistic lesions. Random rotation, horizontal and vertical flips, mild perspective changes, controlled brightness variation, contrast variation, and blur simulation are used. Aggressive color shifts are avoided because they can distort skin tone and lesion pigmentation. For fairness-sensitive training, augmentation is balanced so that underrepresented tone groups receive additional variation without simply duplicating the same images. This reduces overfitting while preserving the natural diversity of lesion presentation.

Image quality validation removes corrupt, duplicate, and unusable images before training. Lesion-centered cropping preserves both the lesion and nearby healthy skin needed for tone estimation.

Tone-aware normalization reduces illumination differences while avoiding loss of diagnostically relevant pigment patterns.

Fairness-focused augmentation improves representation for darker skin tones without creating artificial color artifacts.

5.3 FairDermNet Model Implementation

FairDermNet is implemented as a lightweight convolutional architecture based on MobileNetV3-Large. MobileNetV3 is selected because it provides a strong balance between accuracy and mobile efficiency through depthwise separable convolutions, squeeze-and-excitation blocks, and hardware-aware activation functions. The base network is modified to support two coordinated tasks: lesion risk classification and skin tone group prediction. The skin tone branch is not included to label the user clinically; it is included so the model learns to monitor and correct performance differences across tone groups.

The model receives a 224 by 224 RGB tensor after preprocessing. Feature extraction occurs through the MobileNetV3 backbone, followed by a fairness attention module. The fairness attention module learns tone-informed feature weights that encourage the classifier to attend to lesion morphology rather than background pigmentation alone. A global context aggregator then combines local lesion features with broader color and texture patterns. The final risk head outputs benign and malignant probabilities, while the auxiliary tone head outputs the predicted Fitzpatrick group used during fairness training and evaluation.

Training uses a multi-objective loss. The primary classification loss optimizes lesion risk prediction. The auxiliary tone loss improves the reliability of tone estimation and helps stabilize the fairness branch. A fairness regularization term penalizes large differences in false positive rate, false negative rate, and balanced accuracy across tone groups. The penalty is weighted carefully so that fairness improves without collapsing overall diagnostic performance. Ablation experiments compare the base MobileNetV3, MobileNetV3 with augmentation, and FairDermNet with the complete fairness module.

Model calibration is implemented after training using validation-set confidence analysis. Raw softmax probabilities can be overconfident, especially on out-of-distribution smartphone images. Temperature scaling and threshold tuning are used to align predicted confidence with observed correctness. DermaLens reports risk as Low, Medium, or High instead of presenting an absolute diagnosis. This reduces the chance that a user interprets the result as a substitute for clinical evaluation.

Backbone: MobileNetV3-Large with depthwise separable convolution for efficient mobile inference.

Auxiliary branch: Fitzpatrick group prediction used for fairness-aware training and monitoring.

Regularization: Equalized-odds inspired penalty to reduce disparity in false positive and false negative rates.

Calibration: Temperature scaling and risk-threshold tuning to improve confidence reliability.

5.4 Mobile Application Modules

The Flutter application is organized into modules that correspond to the user workflows described in Chapter 4. The camera module uses the device camera stream, an overlay guide, and quality feedback prompts. The preprocessing module receives a captured frame, checks blur and lighting, performs basic lesion localization, estimates skin tone, and prepares the tensor for the model. The inference module wraps the TensorFlow Lite interpreter and exposes a simple function that returns risk probabilities, predicted tone group, latency, and intermediate metadata.

The results module presents the output in a clinically cautious manner. It displays the risk category, confidence band, heatmap overlay, image quality notes, and recommended next action. High-risk results encourage prompt professional consultation, while medium-risk results recommend monitoring and clinician review if changes are observed. Low-risk results are accompanied by educational reminders because a low score should not encourage users to ignore persistent or changing lesions. The interface avoids diagnostic language such as ’you have melanoma’ and instead uses decision-support wording.

The history module stores scan records locally using an encrypted SQLite database. Each record includes timestamp, lesion nickname, body location, raw score, calibrated score, risk category, heatmap thumbnail, quality notes, and optional user observations such as itching, bleeding, or rapid growth. Longitudinal comparison is implemented by grouping scans under a lesion profile. When a new scan is saved, the system compares risk score trend, color variation, approximate diameter change, and user-reported symptoms against previous records.

The report module generates a clinician-friendly PDF containing the original image, heatmap, risk category, confidence information, ABCDE observations, timeline summary, and disclaimer. The report is intentionally designed for sharing by the user rather than automatic transmission. This respects privacy and keeps the user in control of their own health data.

5.5 Testing Strategy

Testing is divided into unit testing, integration testing, model evaluation, usability testing, and device performance testing. Unit tests cover preprocessing utilities, image quality scoring, risk category conversion, database serialization, and report field generation.
Integration tests cover the complete scan workflow from captured image to saved record. Model evaluation tests are run separately in the training environment and measure accuracy, sensitivity, specificity, balanced accuracy, calibration error, and fairness disparity across skin tone groups.

Device performance testing is essential because DermaLens is designed for offline use on consumer smartphones. The target benchmark device is a Snapdragon 720G-class phone, with additional tests on high-end and entry-level hardware. Measurements include model loading time, average inference latency, heatmap generation time, peak memory usage, battery impact during repeated scans, and application responsiveness. The model must remain small enough for practical app distribution while maintaining clinically meaningful accuracy.

Usability testing evaluates whether users can capture suitable lesion images and interpret results responsibly. The planned study uses the System Usability Scale, task completion rates, capture retry counts, trust ratings, and comprehension questions about the meaning of risk categories and heatmaps. A successful usability outcome requires not only that users like the application, but that they understand its limits and know when to seek medical advice.

Unit tests verify individual functions such as blur detection, score calibration, and database mapping.
Integration tests verify the scan, explain, save, compare, and export workflow.
Model tests measure accuracy and fairness across Fitzpatrick groups I to VI.
Device tests measure latency, memory, startup time, and repeated-use stability.

5.6 Evaluation Metrics

DermaLens is evaluated with both traditional machine learning metrics and deployment-oriented metrics. Accuracy alone is insufficient because the dataset may be imbalanced and because a clinically useful screening tool must pay close attention to false negatives.
Sensitivity measures the proportion of malignant cases correctly identified, while specificity measures the proportion of benign cases correctly identified. Balanced accuracy is used to reduce the effect of class imbalance. The area under the ROC curve is used to examine ranking quality across thresholds.

Fairness is evaluated by comparing performance across skin tone groups. The key fairness metrics are maximum group accuracy gap, false negative rate gap, false positive rate gap, and calibration difference. A model that performs well overall but misses malignant lesions more often in darker skin tones would be unacceptable for the social purpose of this project.
Therefore, the target is to keep the performance gap below five percentage points while preserving strong sensitivity.

Explainability is evaluated qualitatively and quantitatively. Qualitatively, GradCAM++ heatmaps are reviewed to ensure highlighted regions overlap with the lesion rather than irrelevant background, ruler marks, shadows, or hair. Quantitatively, explanation localization can be measured against available segmentation masks using overlap metrics. In the mobile interface, explanations are also evaluated through user comprehension: users should understand that the heatmap shows influential regions, not medically confirmed cancer boundaries.

Operational metrics include model size, inference latency, heatmap generation time, app launch time, and storage growth per scan. These metrics determine whether the system remains practical in low-connectivity and resource-constrained settings. A fairness-aware model that is too large or too slow would fail the project's accessibility goal.

Chapter 6: Risk Management, Ethics, and Future Scope

6.1 Clinical Safety Considerations

DermaLens is positioned as a decision-support and awareness tool rather than a diagnostic system. This distinction is central to clinical safety. The application can help users identify lesions that deserve attention, document changes over time, and prepare useful information for a clinician. It cannot replace dermoscopy, biopsy, histopathology, or professional examination. The interface and generated report must therefore communicate uncertainty clearly and repeatedly without creating unnecessary fear.

The most serious clinical risk is a false negative result, where a malignant lesion is classified as low risk and the user delays care. To reduce this risk, the model prioritizes sensitivity, the application includes educational guidance about warning signs, and the results screen encourages clinical review for persistent, changing, bleeding, painful, or unusual lesions regardless of the AI score. The system also avoids showing low-risk results as a guarantee. A low-risk output means that the model did not detect high-risk patterns in that image under current conditions.

False positives are also important because they can cause anxiety and unnecessary clinic visits. DermaLens addresses this through confidence calibration, medium-risk categorization, image quality feedback, and longitudinal monitoring. Instead of pushing every uncertain lesion into a high-risk warning, the app can advise retaking a poor-quality image, monitoring a lesion over time, or consulting a clinician based on context. This balances caution with usability.

6.2 Privacy and Data Protection

Skin lesion images are sensitive health data. They may reveal body location, skin condition, age-related features, and potentially identifiable marks. DermaLens reduces privacy risk by performing inference on the device and avoiding cloud upload for its core functionality. The application does not require an account for scanning, and no image leaves the device unless the user explicitly exports or shares a report.

Local storage is encrypted at rest. The database stores scan metadata, thumbnails, scores, heatmap references, and lesion timeline information. Encryption keys are managed through platform-secure storage, such as Android Keystore or iOS Keychain. The application should also provide a clear delete function so users can remove individual scans or all local health

data. Report export is treated as a deliberate action because generated PDF files may be shared through external apps that follow their own privacy policies.

Future versions may support optional federated learning or privacy-preserving analytics. However, such features should remain opt-in and should be designed with differential privacy or secure aggregation where appropriate. The current project scope deliberately avoids background upload because offline privacy is one of the main differentiators from existing commercial tools.

6.3 Ethical and Fairness Considerations

Fairness is not a secondary feature in DermaLens; it is a core design requirement. Dermatology datasets have historically underrepresented darker skin tones, and many visual examples used in medical education are biased toward lighter skin. If these biases are transferred into AI systems, the users who already face barriers to dermatological care may receive the least reliable predictions. DermaLens addresses this by measuring performance separately across skin tone groups and by making fairness a training objective rather than only a post-deployment audit.

Ethical deployment also requires careful communication. The application should avoid implying that skin tone categories are fixed biological labels. Fitzpatrick groups are used as practical proxies for evaluating visual model performance, not as a complete description of race, ethnicity, or individual identity. The app should not display sensitive tone labels unnecessarily to the user. Instead, tone estimation can operate internally to improve normalization and fairness monitoring.

Bias may still enter through image quality, device differences, lighting conditions, and regional differences in lesion prevalence. For this reason, fairness evaluation should be continuous. Future datasets should include images from multiple geographies, devices, age groups, and skin tones, with clear consent and annotation standards. Human review by dermatology experts remains important when validating performance and interpreting failure cases.

6.4 Limitations

The first version of DermaLens has several limitations. It focuses primarily on pigmented lesions and melanoma risk rather than the full range of dermatological conditions. Non-pigmented basal cell carcinoma, squamous cell carcinoma, inflammatory conditions, fungal

infections, and rare skin disorders are outside the core scope. The model may also be less reliable when images contain heavy hair occlusion, tattoos, makeup, wounds, shadows, glare, or very small lesions.

Another limitation is dataset labeling. Public datasets often contain labels derived from different clinical workflows, including histopathology, expert consensus, and follow-up. Combining them introduces label noise and domain differences. Smartphone images also vary dramatically across camera sensors and lighting conditions. Although preprocessing and augmentation reduce this variation, they cannot eliminate it completely. Real-world validation with diverse users is required before any deployment beyond a research prototype.

The explainability layer also has limitations. GradCAM++ identifies image regions that influenced the model, but it does not prove that the model is using medically valid reasoning. A heatmap can increase transparency, yet it can also create false confidence if users interpret it as a definitive lesion boundary. The interface must therefore describe explanations cautiously and pair them with actionable guidance.

6.5 Future Enhancements

Future work can extend DermaLens in several directions. The model can be improved with larger, more balanced smartphone image datasets and dermatologist-reviewed annotations. Additional lesion categories can be introduced once sufficient data and validation are available. The app can also support better body-location mapping, reminder scheduling for follow-up scans, and more precise longitudinal measurement using reference objects or camera calibration.

Federated learning is a promising future enhancement because it could allow the model to improve from distributed user data without centralizing raw images. A federated version would need strong privacy safeguards, secure aggregation, bias monitoring, and user consent. Another direction is clinician-assisted review, where users can export structured reports to dermatologists or clinics and receive professional feedback through existing healthcare channels.

The explainability component can also be expanded. In addition to GradCAM++, the app could show ABCDE feature-level summaries, compare current and previous scans side by side, and explain why image quality affects reliability. For accessibility, the interface can support multiple languages, voice guidance during image capture, high-contrast display modes, and simplified result summaries for users with limited health literacy.

Expand disease coverage after collecting validated data for additional lesion categories.

Introduce opt-in federated learning with privacy-preserving safeguards.

Improve longitudinal measurement with body maps, calibrated scale references, and scheduled reminders.

Add multilingual educational content and accessibility-focused capture guidance.

6.6 Conclusion

DermaLens demonstrates how a mobile AI system for skin lesion risk assessment can be designed around privacy, fairness, and explainability from the beginning. The project combines a lightweight on-device model, tone-aware preprocessing, fairness regularization, GradCAM++ visual explanations, local encrypted storage, and longitudinal tracking.

Together, these components address several weaknesses found in existing skin cancer screening applications.

The expanded design also shows that technical performance alone is not enough for responsible health AI. The system must be usable, cautious in its language, transparent about uncertainty, respectful of privacy, and evaluated across the populations it intends to serve.

With further clinical validation and dataset expansion, DermaLens can become a meaningful decision-support tool that helps users seek care earlier while preserving control over their personal health data.

Chapter 7: Deployment Plan and Maintenance

7.1 Deployment Readiness

Deployment readiness for DermaLens requires more than packaging the Flutter application and exporting a TensorFlow Lite model. Because the project handles sensitive health-related images and produces risk guidance, readiness must be assessed across technical, usability, privacy, and documentation dimensions. A release candidate should be considered ready only when the scanning workflow is stable, the model file is versioned, all dependencies are documented, and the user-facing disclaimers have been reviewed for clarity.

The deployment package contains the Flutter mobile application, the quantized FairDermNet model, preprocessing configuration files, educational content assets, and report templates.

Each release should include a model card describing the training data, intended use, limitations, fairness results, and known failure cases. This model card is important because future developers, evaluators, and clinical collaborators need to understand the context in which the model was produced. Without such documentation, a technically successful model can become difficult to audit or responsibly improve.

Before public testing, the app should pass a structured release checklist. This checklist includes installation on supported Android and iOS versions, first-run permission handling, offline scan execution, database encryption verification, report export validation, accessibility checks, crash-free repeated scans, and graceful behavior when the camera or storage permission is denied. The checklist also verifies that no raw images are uploaded during normal scanning and that all exported files are generated only after user action.

Confirm the model version, preprocessing version, and app version are recorded together for every release.

Verify that the application completes the core scan workflow without internet connectivity.

Confirm privacy notices, disclaimers, and educational content are visible at the appropriate points.

Test export reports on multiple devices and PDF readers to ensure consistent formatting.

7.2 Version Control and Reproducibility

Reproducibility is a major concern in machine learning projects because model behavior depends on datasets, preprocessing choices, random seeds, training configuration, and library versions. DermaLens should therefore maintain separate but linked version histories for the mobile application and the model training pipeline. The mobile repository stores Flutter code, app assets, report templates, and integration tests. The model repository stores training scripts, preprocessing notebooks, experiment configurations, exported model artifacts, and evaluation reports.

Every exported model should be assigned a semantic version and a content hash. The app should log the model version used for each scan record so that later review can identify which model produced a given result. This matters when the model is retrained or recalibrated. If a user exports an old report, the clinician should be able to see the model version and understand that scores from different versions may not be directly comparable.

Experiment tracking tools can be used to store training metrics, fairness metrics, hyperparameters, dataset splits, and model artifacts. Even a lightweight tracking spreadsheet or structured JSON log is better than undocumented experiments. The minimum reproducibility package should include the dataset split seed, augmentation settings, loss weights, optimizer settings, number of epochs, best checkpoint criteria, calibration method, and threshold selection procedure.

The report should also include a reproducibility note for academic evaluation. This note clarifies which parts of the system are implemented in the current prototype, which parts are planned, and how future contributors can recreate the evaluation. Clear reproducibility practices improve trust because they make it easier for others to verify claims rather than relying only on summary results.

7.3 Maintenance Strategy

Maintenance for DermaLens includes application maintenance, model maintenance, content maintenance, and privacy maintenance. Application maintenance covers bug fixes, operating system compatibility, dependency updates, camera plugin changes, and performance improvements. Model maintenance covers new datasets, retraining, fairness audits, calibration updates, and validation on new device image distributions. Content maintenance covers educational text, disclaimers, report wording, and clinical guidance links. Privacy maintenance covers encryption behavior, permission changes, export handling, and deletion workflows.

A periodic maintenance cycle is recommended. During each cycle, developers should review crash reports from test builds, evaluate user feedback, run regression tests, and re-check model performance on a holdout dataset. Any model update should be treated as a significant release rather than a silent asset replacement. This is because even small model changes can alter risk categories and user behavior. A model update should therefore include updated metrics and a short explanation of why the change was made.

Dependency maintenance is especially important in Flutter and mobile ML projects. Camera APIs, TensorFlow Lite plugins, Android SDK versions, and iOS permission requirements can change over time. Dependency updates should be tested on the benchmark device set before release. The project should avoid unnecessary dependency growth because each additional package increases maintenance burden, security surface, and potential incompatibility.

Maintenance also includes retiring unsafe features. If a future audit shows that a feature causes misunderstanding, excessive anxiety, privacy risk, or unfair performance, it should be revised or removed. Responsible health software must be willing to change based on evidence rather than preserving features only because they are technically impressive.

7.4 User Education Plan

User education is necessary because DermaLens operates in a sensitive area where misunderstanding can have serious consequences. The app should teach users how to capture better images, how to understand the risk categories, what the heatmap does and does not mean, and when to consult a clinician. This education should be placed inside the workflow rather than hidden in a long help document. Short contextual messages are more likely to be read at the moment they matter.

The capture screen should educate through feedback. If the image is blurry, the app should say that blur can reduce reliability and ask the user to retake the image. If lighting is poor, the app should recommend moving to diffuse natural light or avoiding harsh shadows. If the lesion is too small in the frame, the overlay can guide the user to move closer while keeping focus. These messages improve data quality and teach the user why capture conditions matter.

The results screen should educate through careful wording. Low risk should be described as ’no high-risk visual pattern detected in this image’ rather than ’safe.’ Medium risk should explain uncertainty and recommend monitoring or clinical review based on symptoms and change. High risk should recommend prompt professional consultation without declaring a

diagnosis. Heatmaps should be explained as model attention maps, not as medical proof of malignancy.

The educational content section can include the ABCDE rule, examples of concerning changes, guidance on photographing lesions over time, and instructions for preparing for a dermatology appointment. For accessibility, educational text should be written in plain language, translated in future versions, and supported by high-contrast visuals. The content should also remind users that lesions on nails, palms, soles, mucosal areas, or rapidly changing lesions deserve professional attention even if the app result appears low risk.

7.5 Quality Assurance Checklist

Quality assurance combines automated tests with manual review. Automated tests are useful for repeatability, but manual testing remains important because camera behavior, visual explanations, and health communication depend heavily on real user interaction. The QA process should therefore include scripted scan scenarios and exploratory testing. Testers should intentionally capture images under difficult conditions to verify that the app refuses or warns about unreliable inputs.

A representative QA checklist includes permission testing, capture testing, preprocessing testing, inference testing, result display testing, history testing, report export testing, deletion testing, and offline testing. Each checklist item should include expected behavior and failure handling. For example, if the model file is missing or corrupted, the app should not crash; it should show a clear error and prevent risk scoring until the issue is fixed. If storage is full, the app should complete the scan display but warn that the record cannot be saved.

Fairness QA requires deliberate test coverage across skin tones and lighting conditions. The test set should include examples from all Fitzpatrick groups, multiple device cameras, and varied backgrounds. Testers should check whether the app repeatedly asks users with darker skin tones to retake images under conditions that would pass for lighter skin tones. Such capture-stage bias would undermine the fairness goal even if the model itself performed well in offline evaluation.

Permission denied: app explains why camera or storage permission is needed and offers a recovery path.

Poor image quality: app prompts retake instead of producing unreliable confidence.

Offline mode: scan, heatmap generation, saving, and report generation work without network access.

Data deletion: individual scans and full local history can be removed by the user. Fairness review: capture quality and risk scoring are checked across diverse skin tones.

7.6 Expected Impact

The expected impact of DermaLens is strongest in early awareness, documentation, and care-seeking support. The application can help users notice suspicious lesions earlier, store a visual timeline, and approach clinicians with organized information. In regions with limited dermatology access, even a cautious decision-support tool may encourage users to seek care sooner for lesions that are changing or high risk. The offline design also makes the tool more suitable for low-connectivity environments.

The project also has educational impact. By teaching users about image quality, lesion changes, and the ABCDE rule, DermaLens can improve skin health literacy. The heatmap feature may help users understand that AI predictions are based on image patterns, while the report export feature can improve communication between users and clinicians. These benefits remain valuable even when the AI score is treated cautiously.

From a research perspective, DermaLens contributes a design pattern for fairness-aware mobile medical AI. It demonstrates that fairness, privacy, explainability, and mobile efficiency can be integrated into a single system architecture rather than addressed as separate afterthoughts. The project can be extended into future academic work on tone-aware preprocessing, lightweight explainable models, longitudinal lesion tracking, and human-centered evaluation of medical AI interfaces.

7.7 Summary

The deployment and maintenance plan completes the technical design by describing how DermaLens can remain reliable after implementation. A responsible release requires versioned model artifacts, reproducible training records, privacy-preserving defaults, clear user education, structured QA, and continued fairness monitoring. These practices help ensure that the system remains aligned with its original goals rather than becoming only a one-time prototype.

By combining careful deployment planning with the architecture, modeling, implementation, and ethical analysis described in earlier chapters, DermaLens becomes a more complete project. It is not only an AI model and not only a mobile app. It is a full decision-support workflow designed around real constraints: diverse users, imperfect smartphone images,

privacy concerns, limited connectivity, and the need for trustworthy communication in health-related contexts.

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